Diversity Consistency Learning for Remote-Sensing Object Recognition With Limited Labels

Wenda Zhao, *Member, IEEE*, Tingting Tong, Haipeng Wang, Fan Zhao, You He, and Huchuan Lu, *Senior Member, IEEE*

***Abstract*— Annotating remote-sensing object recognition needs high professionalism, and thus limited labeled samples are available. Suffering from this, general remote-sensing object recognition methods are facing low recognition accuracy. Addressing this issue, this article proposes a diversity consistency learning (DCL) for remote-sensing object recognition with limited labels. Specifically, the diversity generation model (DGM) is designed as a teacher model to generate diverse results, which is trained with labeled samples. Then, a round consistency distillation model (RCDM) is introduced to distill the knowledge of diverse pseudo-labels to a student network, which is trained with unlabeled samples. Especially, diverse pseudo-labels are generated by the well-trained DGM, which can improve recognition accuracy since diverse pseudo-label errors can cancel each other out. Extensive experiments on two widely used datasets of FS23 and HRSC2016 demonstrate the superior performance of our method compared with the state of the arts (SOTAs).**

***Index Terms*— Diversity consistency learning (DCL), limited labeled samples, remote-sensing object recognition.**

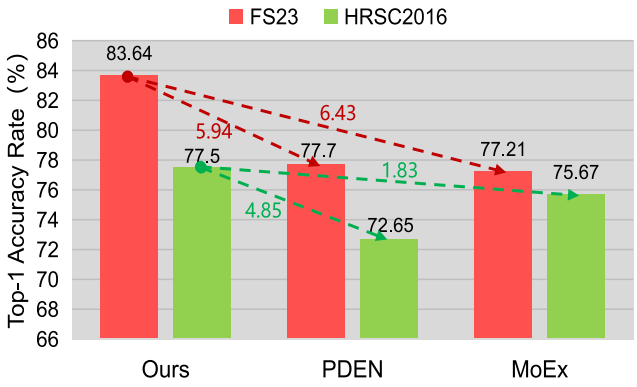


Fig. 1. Performance comparison of different methods on the FS23 dataset [7] and the HRSC2016 dataset [9]. Our method achieves superior performance against state-of-the-art methods of PDEN [10] and MoEx [11].

I. Introduction

REMOTE-sensing object recognition based on deep learning [1], [2], [3], [4], [5], [6] has become an important research topic for better scene interpretation. However, existing object recognition frameworks are trained that rely on large-scale labeled datasets, which is unavailability in practice. For instance, the fine-grained remote-sensing ship classification datasets [1], [7], [8] need highly professional and time-consuming annotation, thus containing limited labeled samples. Unfortunately, general remote-sensing object recognition methods suffer from the limited labeled datasets, thereby resulting in low recognition accuracy.

Manuscript received 1 July 2022; revised 15 August 2022; accepted 9 September 2022. Date of publication 30 September 2022; date of current version 18 October 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62176038, Grant 62076249, Grant 62001450, and Grant U1903215; and in part by the Science and Technology Star of Dalian under Grant 2021RQ054 and Grant 2021RQ059. *(Corresponding author: Fan Zhao.)*

Wenda Zhao, Tingting Tong, and Huchuan Lu are with the Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of the Ministry of Education and the School of Information and Communication Engineering, Dalian University of Technology, Dalian 116024, China (e-mail: zhaowenda@dlut.edu.cn; ttt31909076@mail.dlut.edu.cn; lhchuan@dlut.edu.cn).

Haipeng Wang and You He are with the Research Institute of Information Fusion, Naval Aviation University, Yantai 264001, China (e-mail: whp5691@163.com; heyou_f@126.com).

Fan Zhao is with the School of Physics and Electronic Technology, Liaoning Normal University, Dalian 116029, China (e-mail: fan_zhao20@163.com).

Digital Object Identifier 10.1109/TGRS.2022.3210980

Semisupervised learning is the prominent line for relieving the issue of limited labels by utilizing large-scale unlabeled samples. It has been explored in image classification [12], [13], [14], [15], semantic segmentation [16], [17], [18], [19], object detection [20], [21], [22], [23], [24], [25], and so on. Semisupervised methods are mainly divided into two categories: pseudo-label methods [12], [15], [22], [23], [25] and consistency regularization methods [16], [17], [19]. On the one hand, pseudo-label methods aim to generate pseudo-truths by using a pretrained network for unlabeled data, and then the generated pseudo-truths and the labeled data are jointly utilized to train the network. However, partial inaccurate pseudo-truths are not conducive to the prediction performance of the network. On the other hand, consistency regularization methods constrain the consistency of network outputs that are generated from different-perturbation unlabeled data, and thus the network feature extraction capability is enhanced. However, consistency regularization methods rely on the effectiveness of data perturbation.

Different from the above methods, we relieve the challenge of limited labeled data by the proposed diversity consistency learning (DCL) for remote object recognition. Specifically, DCL consists of a diversity generation model (DGM) and a round-consistency distillation model (RCDM). DGM focuses on generating diverse pseudo-labels with labeled samples. Since diverse pseudo-label errors can cancel each other out, which improves the prediction accuracy. Sequentially,

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

RCDM distills diverse pseudo-labels to a student model in a round-consistency manner with unlabeled samples. As a result, the student model will generate high-precision prediction results, thereby alleviating the limited label issue.

More specifically, given training datasets of labeled samples D_L and unlabeled samples D_U, our model is trained by two stages. In the first stage, DGM is designed as a teacher model, which is trained with labeled samples D_L for generating diverse pseudo-labels. Structurally, DGM utilizes an encoder for feature extraction and is followed by multiple classification heads $\{h_1, h_2, \ldots, h_M\}$ for pseudo-label predictions. Especially, feature diversity loss is proposed to enhance the diversity of the generated pseudo-labels. In the second stage, RCDM is implemented to distill the knowledge of diverse pseudo-labels to the student network, where the student network has the same encoder and one classification head as the teacher model but with different parameters. Different from existing distillation methods [26], [27], [28], [29], the pseudo-labels of D_U generated by multiple classification heads $\{h_1, h_2, \ldots, h_M\}$ in DGM are distilled to the student model in the round-consistency way. Hence, the student model can get high-precision prediction results under limited labeled data. As shown in Fig. 1, comparing with state of the art (SOTA) methods of progressive domain expansion network (PDEN) [10] and moment exchange (MoEx) [11], our method improves the top-1 accuracy rate of 5.94%&6.43% and 4.85%&1.83% on FS23 and HRSC2016 datasets, respectively.

The main contributions are summarized as follows.

1) We attempt to study remote-sensing object recognition with limited labels and propose a DCL framework.
2) DGM is designed to generate the diverse pseudo-labels with limited labeled samples and then the round consistency distillation (RCD) is implemented to distill diverse pseudo-labels to the student model with unlabeled samples, thereby improving prediction precision since diverse pseudo-label errors can cancel each other out.
3) Extensive experiments on two widely used datasets demonstrate the effectiveness of our method compared with the SOTAs.

II. Related Work

A. Fully-Supervised Method

Fully-supervised object recognition models have made progress. For example, Mackowiak et al. [30] introduce generative classifiers to enhance the interpretability and robustness of the object recognition framework. Ji et al. [31] design attention convolutional binary neural tree to capture multilevel features. Wang et al. [32] explore supervised contrastive learning strategy for better image representations from imbalanced data, thereby boosting the recognition accuracy.

Recently, many pioneering works for remote-sensing object recognition have been proposed. Zhao et al. [1] propose a feature balance strategy to improve low-resolution remote-sensing object recognition and build a low-resolution fine-grained object (LFS) dataset for evaluation. Besides, Zhao et al. [2] improve long-tailed object recognition in aerial images by the proposed hierarchical distillation method. Wang et al. [6] adopt an automatic search strategy to find the appropriate network architecture for remote-sensing object recognition. Ma et al. [33] propose a global and local consistent network for mining useful information from remote-sensing images, thus improving the object recognition accuracy. [34] aims to enhance the scale information and reserve context information for multiclass ship recognition. Chen and Qian [35] focus on hierarchical multilabel ship recognition and design a novel loss function.

Overall, fully-supervised approaches deliver excellent performance by utilizing lots of labeled samples. However, the limited labeled samples in a remote-sensing environment may reduce model accuracy.

B. Semisupervised Method

Semisupervised methods can utilize the limited labeled samples together with a larger-scale unlabeled sample to train models. Various semisupervised methods have been proposed, for example, pseudo-label methods [12], [15], [22], [23], [25], [36], [37] and consistency regularization methods [16], [17], [19], [38], [39], [40], [41]. Pseudo-label methods first generate pseudo-truth for unlabeled samples and then the generated pseudo-truth and the labeled samples are jointly utilized to train the final model. Lee et al. [37] propose to train the network with labeled and unlabeled samples simultaneously. Zhou et al. [25] use an instant pseudo-label with extended weak–strong data augmentations to train the final model. Consistency regularization methods constrain the consistency of network outputs generated by different-perturbation unlabeled samples. Thus, many works focus on the efficient perturbation sample generation, such as data augmentation through random disturbances [38], [39], [40] and data disturbance increasing [41]. Besides, Tarvainen and Valpola [42] propose a mean teacher model that averages model weights instead of label predictions.

Pseudo-label methods can improve network performance, albeit with limited labels. However, the model performance often suffers from partial inaccurate pseudo-truth when only one pseudo-label is generated for each input. Therefore, we propose the DGM to generate the diverse pseudo-labels, and thus diverse pseudo-label errors can cancel each other out which improve the prediction accuracy.

C. Diversity-Learning Method

Diversity learning aims to generate diverse information that enhances the model accuracy by taking advantage of the complementary information. Generally, diversity learning methods can be divided into two categories of multiple network branch methods and multiple head methods. For example, varied multiple branch strategies [28], [29], [43], [44], [45] are proposed to improve the diversity. Works [46], [47], [48] are typical multiple head branch methods.

Existing diversity learning methods intend to integrate diverse information in an unsophisticated way, such as regarding the weighted average of diverse information. However, such an integration strategy may have a certain degree of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

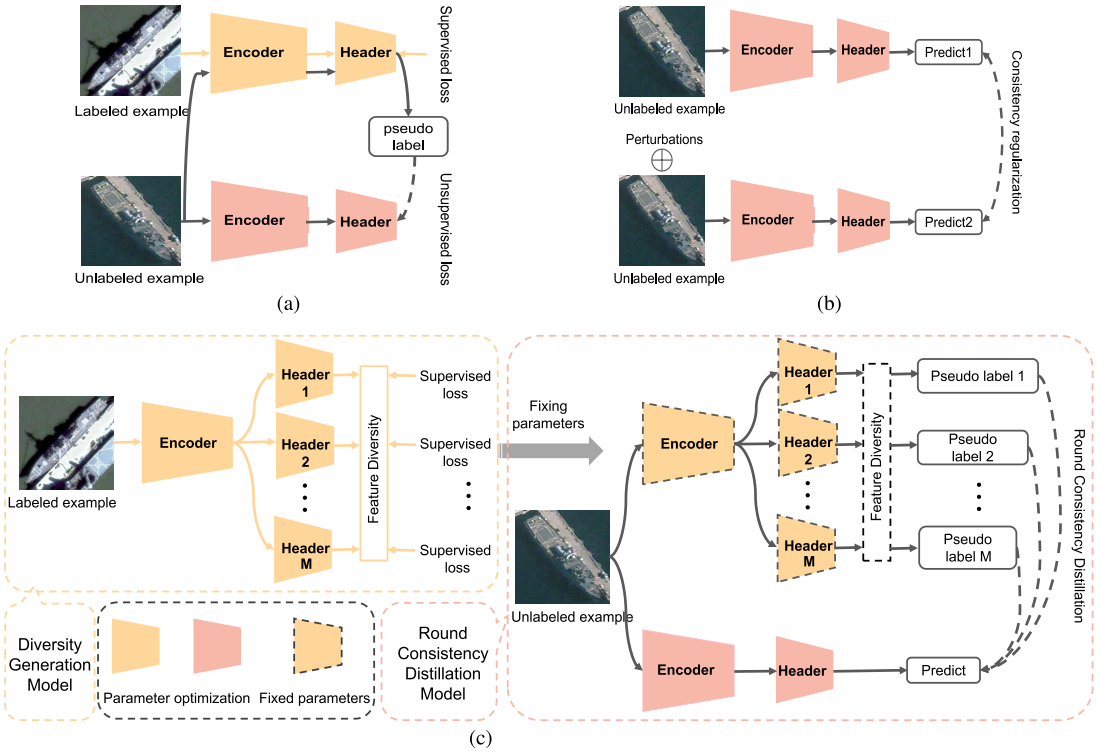


Fig. 2. Motivation of our DCL framework. (a) Pseudo-label method generates one pseudo-label for each unlabeled example, which is sensitive to noisy pseudo-label. (b) Consistency regularization method uses a relatively random perturbation operation, which lacks task pertinence. Therefore, we first train DGM with little labeled samples in a supervised manner and then use it to generate diverse pseudo-labels for each unlabeled sample. (c) RCDM is proposed to distill diverse pseudo-label information to the final model with the help of large-scale unlabeled samples.

destruction to the diversity. In this article, we first use multiple head branches to generate diverse information and then propose the RCDM that distills diverse information to the student model in a round consistency manner. Thus, diverse information can be fully injected into the student model for obtaining accurate object recognition results.

III. Diversity Consistency Learning

A. Motivation and Framework

Due to the limited labeled samples, remote-sensing object recognition performance encounters low accuracy. Semisupervised learning is a prominent line for relieving this challenge, where pseudo-label methods and consistency regularization methods are representative works. As illustrated in Fig. 2(a), the pseudo-label method first trains an auxiliary model with labeled samples and then uses the model to generate pseudo-truth for each input unlabeled sample to train the main model. In consideration of only one pseudo-label generated for each input, the generated pseudo-label lacks diversity. Moreover, the network performance is vulnerable to the partially inaccurate pseudo-truth. As revealed in Fig. 2(b), the consistency regularization method constrains the consistency of the outputs generated by the unlabeled sample and its perturbation. However, the perturbation operation is relatively random, which may limit the performance.

Based on the above discussion, we propose the DCL framework, which includes DGM and RCDM, as shown in Fig. 2(c).

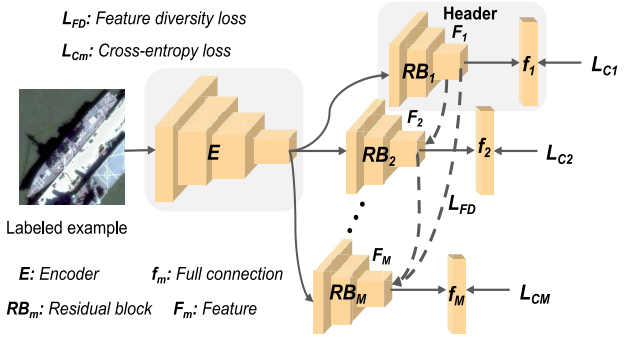


Fig. 3. Structure illustration of the proposed DGM. The encoder E extracts deep features and then feeds them to the multiple classification headers [combined with the residual block (RB_m) and fully connected layer (f_m)].

DGM contains an encoder structure combining multiclassification headers that are trained with little labeled samples in a supervised manner. To enhance the header diversity, a negative correlation feature constraint is built. Second, we propose the RCDM that uses diverse complementary information to improve recognition accuracy. Specifically, DGM is used to generate diverse pseudo-labels for each unlabeled sample, and then RCD is designed to distill diverse pseudo-label information to the student model with the help of large-scale unlabeled samples. More details are introduced as follows.

B. Diversity Generation Model

The structure of DGM is shown in Fig. 3, which is trained using small-scale labeled samples D_L =

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

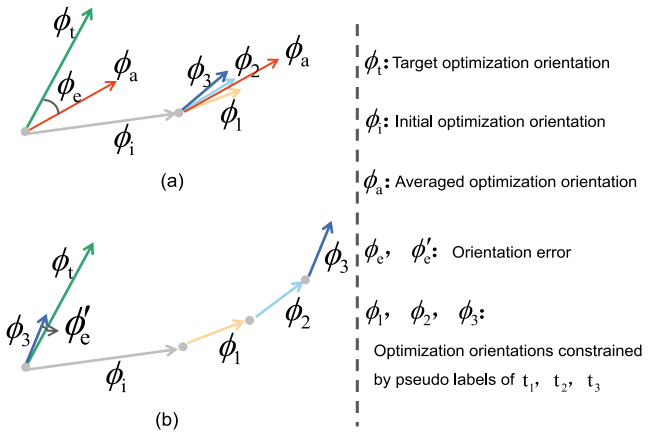


Fig. 4. Optimization orientation illustrations of (a) MCD and (b) RCD.

$\{(d_1^l, g_1), \ldots, (d_N^l, g_N)\}$, where d_n^l represents the nth labeled sample with spatial dimension of $H \times W$, g_n is the label of d_n^l, and $g_n \in R^{C\times 1}$ has C categories. Specifically, DGM is constituted of an encoder E and multiple classification headers $\{H_1, H_2, \ldots, H_M\}$, where M represents the classification header number. E adopts the first three residual blocks of Resnet50 [49], and H_m uses the fourth residual block (RB_m) and fully connected layer (f_m).

Sequentially, we design feature diversity loss L_{FD} to enhance the diversity of classification headers, that is, we constraint the features $\{F_1, F_2, \ldots, F_M\}$ generated by $\{RB_1, RB_2, \ldots, RB_M\}$ to be discrete as follows:

$$L_{FD} = \sum_{i=1}^{M}\sum_{j=1}^{M} \cos\left(\frac{F_i F_j}{|F_i||F_j|}\right)_{(i\neq j)}. \tag{1}$$

Besides, cross-entropy loss L_{C_n} is utilized to supervise the prediction outputs $p_1, p_2, \ldots, p_M$ of multiply classification headers for the nth sample, which is defined as

$$L_{C_n} = -\sum_{m=1}^{M} p_m \log g_m \tag{2}$$

where m represents the mth classification header of the nth sample. g_m denotes the ground truth of the mth classification header.

Thus, the overall loss L_{DGM} for training DGM is formulated as

$$L_{\mathrm{DGM}} = \sum_{n=1}^{N}\left(\omega L_{FD_n} + L_{C_n}\right) \tag{3}$$

where N is the number of samples, and ω is a weight to adjust feature diversity.

C. Round Consistency Distillation Model

Now, we can acquire M predictions by feeding an unlabeled sample into the well-trained DGM. Thus, how to effectively take advantage of the M predictions for the final prediction output remains a challenge to be relieved.

Different from the DGM that is trained over the labeled examples, RCDM is trained using large-scale unlabeled data $D_U = d_1^u, d_2^u, \ldots, d_K^u$. Note that the number of samples K in D_U is much larger than the number of labeled samples N in D_L ($K \gg N$). For simplicity, we utilize an unlabeled sample d_k^u to clarify the RCD procedure. Given d_k^u, the pretrained DGM can generate M pseudo-truths $\{t_1^k, t_2^k, \ldots, t_M^k\}$ by multiply headers $\{H_1, H_2, \ldots, H_M\}$. Subsequently, how to deliver pseudo-truths $\{t_1^k, t_2^k, \ldots, t_M^k\}$ to the student model for generating the final prediction output O_k remains the key.

A native idea is to use mean consistency distillation (MCD). First, the pseudo-truths $\{t_1^k, t_2^k, \ldots, t_M^k\}$ are averaged by

$$\overline{t^k} = \frac{1}{M}\sum_{m=1}^{M} t_m^k. \tag{4}$$

Then, the cross-entropy constraint is used to measure the discrepancy between the pseudo-truth $\overline{p'}$ and expected prediction O_k as follows:

$$L_{\mathrm{MCD}} = -\sum_{k\in B} O_k \log \overline{t^k} \tag{5}$$

where B denotes one batch, and L_{MCD} is the MCD loss.

Algorithm 1 Optimization Process of Our Framework

Input: Labeled dataset D_L and unlabeled dataset D_U.
Stage 1: Training diversity generation model with D_L.
1. Optimizing diversity generation model using Equation (3).
Stage 2: Training round consistency distillation model with D_U.
2. Generating diverse pseudo labels by the well-trained diversity generation model.
3. Optimizing round consistency distillation model using Equation (6).
Output: Parameters of the student model.

Fig. 4(a) shows the optimization procedure of the MCD. Taking $M = 3$, for example, the final optimization orientation ϕ_a is determined by the average of ϕ_1, ϕ_2, and ϕ_3, where ϕ_1, ϕ_2, and ϕ_3 correspond to the optimization orientation constrained by pseudo-labels of t_1^k, t_2^k, and t_3^k. However, such one-step optimization may not achieve the optimal optimization orientation (i.e., a large orientation error ϕ_e).

Therefore, we further design the RCD elaborately to continuously optimize the student network with pseudo-truths $\{t_1^k, t_2^k, \ldots, t_M^k\}$. Specifically, the well-trained DGM first generates M pseudo-labels for an unlabeled sample, and then M pseudo-labels are used to supervise a student model in a round-consistency manner, which is formulated as follows:

$$\begin{aligned} L_{\mathrm{RCD}}(\phi_1) &= -\sum_{k\in B} O_k \log t_1^k \rightarrow \\ L_{\mathrm{RCD}}(\phi_2) &= -\sum_{k\in B} O_k \log t_2^k \rightarrow \\ &\cdots \rightarrow \\ L_{\mathrm{RCD}}(\phi_M) &= -\sum_{k\in B} O_k \log t_M^k \end{aligned} \tag{6}$$

where L_{RCD} is the RCD loss, and $\rightarrow$ stands for the optimization order, which points to the next optimization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

TABLE I

STATISTICAL DETAILS OF THE FS23 DATASET. "TRAINING(L)" EXPRESSES "TRAINING WITH LABELED SAMPLES," AND "TRAINING(UL)" EXPRESSES "TRAINING WITH UNLABELED SAMPLES"

Name	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22
Category	Non-ship	Aircraft-carrier	Destoryer	Landing-craft	frigate	Amphibious-transport-dock	Cruiser	Tarawa-class-amphibious-assult-ship	Amphibious-assult-ship	Command-ship	Submarine	Medical-ship	Combat-boat	Auxiliary-ship	Container-ship	Car-carrier	Hover-craft	Bulk-carrier	Oil-tanker	Fishing-boat	Passenger-ship	Liquefied-gas-ship	barge
Training(L)	128	43	144	28	78	23	77	23	40	23	63	5	37	59	26	19	31	91	43	27	23	24	14
Training(UL)	387	132	434	86	236	72	234	70	123	71	190	17	114	180	81	58	96	274	132	82	70	74	43
Testing	97	34	108	22	59	18	59	18	31	18	48	10	29	45	20	14	24	69	33	20	18	20	11

TABLE II

STATISTICAL DETAILS OF THE HRSC2016 DATASET. "TRAINING(L)" EXPRESSES "TRAINING WITH LABELED SAMPLES," AND "TRAINING(UL)" EXPRESSES "TRAINING WITH UNLABELED SAMPLES"

Name	H0	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16	H17	H18	H19	H20	H21	H22	H23
Category	Ship	Warcraft	Nimitz-class-aircraft-carrier	Enterprise-class-aircraft-carrier	Arleigh-Burke-class-destroyers	Whidbey Island-class-landing-craft	Perry-class-frigate	Sanantonio-class-amphibious-transport-dock	Ticonderago-class-cruiser	Admiiral-Kuznetsov-aircraft-carrier	Austen-class-amphibious-transport-dock	Taraiwa-class-amphibious-assault-ship	Container-ship	Oxo	Car-carrier	Hovercraft	Yacht	Cargo-ship	Cruise-ship	Sumarine	Lute	Medical-ship	Transport-ship	Midway-class-aircraft-carrier
Training(L)	91	12	11	10	60	15	33	13	35	5	12	21	15	11	5	12	5	40	6	12	3	11	16	11
Training(UL)	276	24	36	15	183	46	100	21	106	5	38	66	19	36	5	39	5	122	6	26	3	11	21	11
Testing	167	13	23	10	124	41	111	17	94	2	32	38	15	33	3	44	1	88	4	36	2	10	13	8

Structurally, the student model is composed of an encoder E_S followed by a header H_S. E_S and H_S adopt the same structure as E and H_m of the DGM, respectively, as shown in Fig. 2. As illustrated in Fig. 4(b), optimization orientations ϕ_1, ϕ_2, and ϕ_3 gradually approach the target optimization orientation, thereby obtaining a small orientation error ϕ'_e.

D. Optimizing

The optimization process of our framework includes two stages, as introduced in Algorithm 1. The first stage is to train the DGM with a small-scale labeled sample dataset D_L, and L_{DGM} is used as the loss function. The second stage is to train the RCDM using a large-scale unlabeled dataset D_U. First, diverse pseudo-labels are generated by the well-trained DGM in the first stage. Second, diverse pseudo-labels are delivered to the student model for generating the final prediction output by the RCD [L_{RCD} in (6)].

IV. EXPERIMENT

A. Configuration

1) Dataset: Two widely used remote-sensing classification datasets of FS23 [7] and HRSC2016 [9] are utilized for our experiment. FS23 includes 23 categories, and HRSC2016 includes 24 categories. We use a small number of labeled

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

TABLE III

NUMBER OF STUDY OF CLASSIFICATION HEADERS IN DGM ON THE FS23 DATASET. "DGM_N" STANDS FOR DGM WITH *N* CLASSIFICATION HEADERS. THE OVERALL ACCURACY (OA) AND EACH CLASS ACCURACY ARE LISTED

Set	OA	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22
DGM_1	81.7	79.4	88.2	85.2	72.7	89.8	**77.8**	83.1	72.2	93.5	61.1	93.8	90.0	82.8	71.1	80.0	**100.0**	**100.0**	71.0	81.8	**45.0**	55.6	**100.0**	**100.0**
DGM_2	82.7	87.6	91.2	82.4	75.0	93.2	58.3	**89.8**	88.9	85.4	**91.6**	**94.8**	**95.0**	81.0	60.0	80.0	**100.0**	**100.0**	66.7	**84.8**	30.0	**72.3**	**100.0**	**100.0**
DGM_3	83.1	86.6	90.2	84.9	74.2	**93.8**	59.3	86.9	**90.7**	95.7	88.9	93.1	86.7	82.7	60.0	81.7	83.3	**100.0**	**79.2**	71.7	35.0	68.5	93.3	**100.0**
DGM_4	**84.3**	**92.5**	**91.9**	**87.2**	**86.4**	91.1	55.6	85.1	81.9	**99.2**	81.9	92.7	90.0	**83.6**	**73.9**	**85.0**	82.1	**100.0**	68.8	75.7	43.7	62.5	**100.0**	**100.0**

TABLE IV

EFFECT STUDY OF FEATURE DIVERSITY BY SETTING DIFFERENT ω ON THE FS23 DATASET. THE OA AND EACH CLASS ACCURACY ARE LISTED

ω	OA	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22
0.01	82.9	**88.7**	87.2	85.8	75.8	93.2	**64.8**	86.4	81.5	94.6	83.3	86.1	**100.0**	**91.9**	62.9	**85.0**	85.7	**100.0**	76.3	76.8	23.3	55.6	95.0	**100.0**
0.10	83.0	84.5	**92.2**	**87.0**	**81.8**	91.5	51.9	78.0	77.8	90.3	83.3	**100.0**	76.7	86.2	69.8	75.0	**88.1**	**100.0**	72.5	**86.9**	**45.0**	61.1	**100.0**	**100.0**
1.00	**83.1**	86.6	90.2	84.9	74.2	**93.8**	59.3	**86.9**	**90.7**	**95.7**	**88.9**	93.1	86.7	82.7	60.0	81.7	83.3	**100.0**	**79.2**	71.7	35.0	**68.5**	93.3	**100.0**
5.00	80.7	79.7	85.3	86.4	77.3	89.8	59.3	80.8	68.5	94.6	83.3	95.1	23.3	82.8	**72.6**	**85.0**	83.3	**100.0**	76.3	75.8	31.7	51.8	95.0	**100.0**

TABLE V

RESULTS OF MCD AND RCD ON THE FS23 DATASET. THE OA AND EACH CLASS ACCURACY ARE LISTED

Method	OA	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22
MCD	83.0	**89.7**	**88.2**	**84.3**	**77.3**	**93.2**	66.7	**86.4**	**100.0**	**96.8**	**88.9**	93.8	**70.0**	**86.2**	55.6	**75.0**	**85.7**	**100.0**	72.5	**78.8**	30.0	**66.7**	**100.0**	**100.0**
RCD	**83.6**	83.5	85.3	**84.3**	**77.3**	**93.2**	**77.8**	**86.4**	**100.0**	**96.8**	**88.9**	**97.9**	60.0	82.8	**60.0**	**75.0**	**85.7**	**100.0**	**87.0**	72.7	**35.0**	**66.7**	95.0	**100.0**

samples and a large number of unlabeled samples to train our model. Specifically, 1069 labeled training samples, 3256 unlabeled training samples, and 825 testing samples are selected from FS23. Four hundred and sixty-five labeled training samples, 929 unlabeled training samples, and 1220 testing samples are adopted from FS23. Their statistical details are shown in Tables I and II.

2) Implementation: The experiment is conducted on a GTX 1080Ti GPU using PyTorch. The SG optimizer with a momentum of 0.9 is used. The batch size is set to 8, and input images are 224 $\times$ 224 pixels. The DGM is initialized with the parameters of Resnet50, and the initial weight decay is set to 1e$-$3 and is reduced to 1/10 at every 30 epochs. The total training epoch is 100. The weight decay is set to 1e$-$4 and the total training epoch is 100 for the RCDM.

B. Ablation Study

1) Number of Classification Headers: DGM builds multiple classification headers to generate diverse results, thereby improving object recognition accuracy. Here, we study the number of classification headers. The experiment is built on the FS23 dataset, and the labeled training samples are used to train the DGM equipped with a different number of classification headers. The results are shown in Table III. As can be seen that more classification headers can achieve better performance. Comprehensively considering performance and storage, we select the DGM with three classification headers (i.e., DGM_3) for the experiment.

2) Effect of Feature Diversity: In the stage of training DGM, we add feature diversity loss L_{FD} to enhance the diversity of classification headers. Thus, in (3), we use a weight ω to adjust feature diversity, and a larger ω will make the model generate bigger feature diversity. Here, we discuss the effect of feature diversity by setting different ω. It can be seen from Table IV that the classification accuracy increases when ω becomes larger, for example, $\omega = 1.00$ achieves the top accuracy of 83.1%. However, the classification accuracy decreases when $\omega = 5.00$. The reason may be that too much feature diversity will contain noise, thereby affecting the classification performance.

3) Influence of Round Consistency Distillation: In Section III-C, we introduce two methods of MCD and RCD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

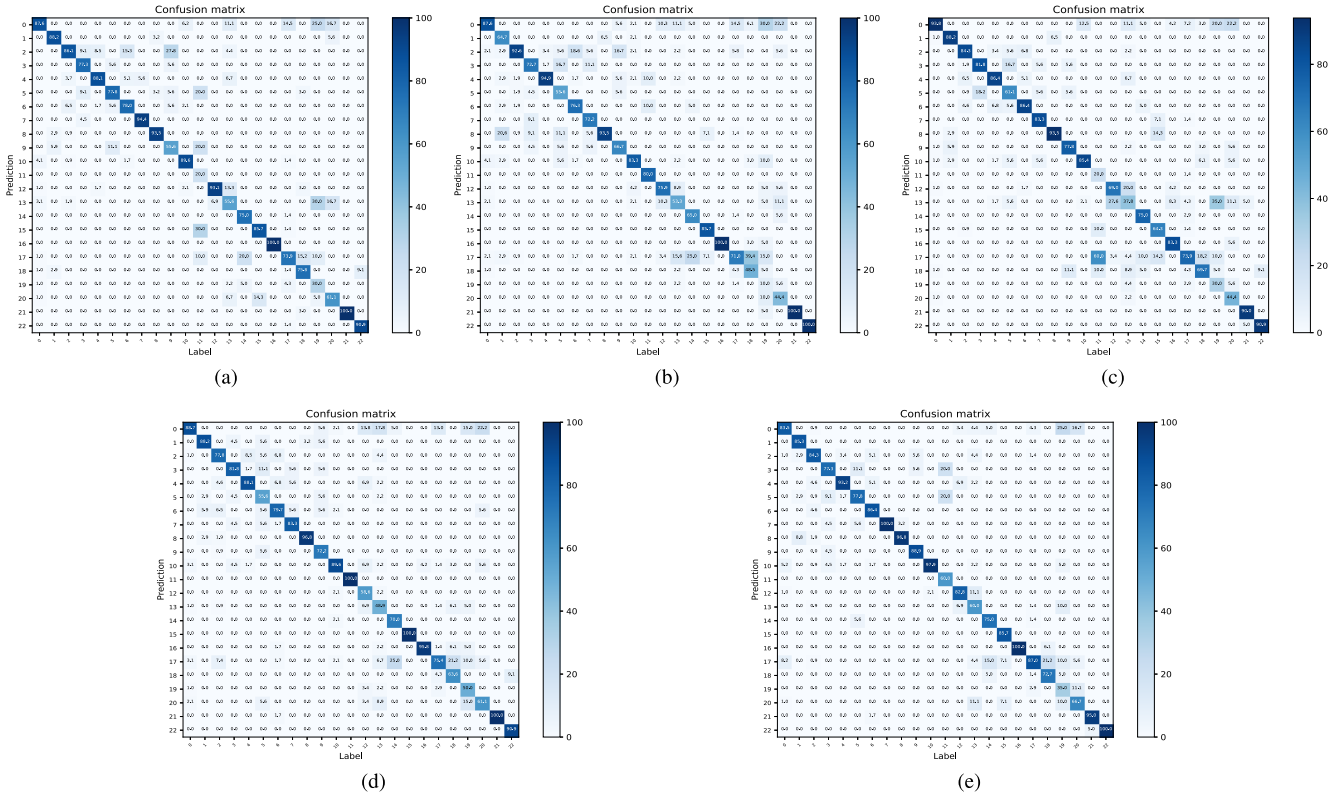


Fig. 5. Confusion matrices of different methods on FS23. The number in the confusion matrix represents the probability that the column's category is predicted as the row's category. (a) LTD. (b) MoEx. (c) PDEN. (d) PMG. (e) Ours.

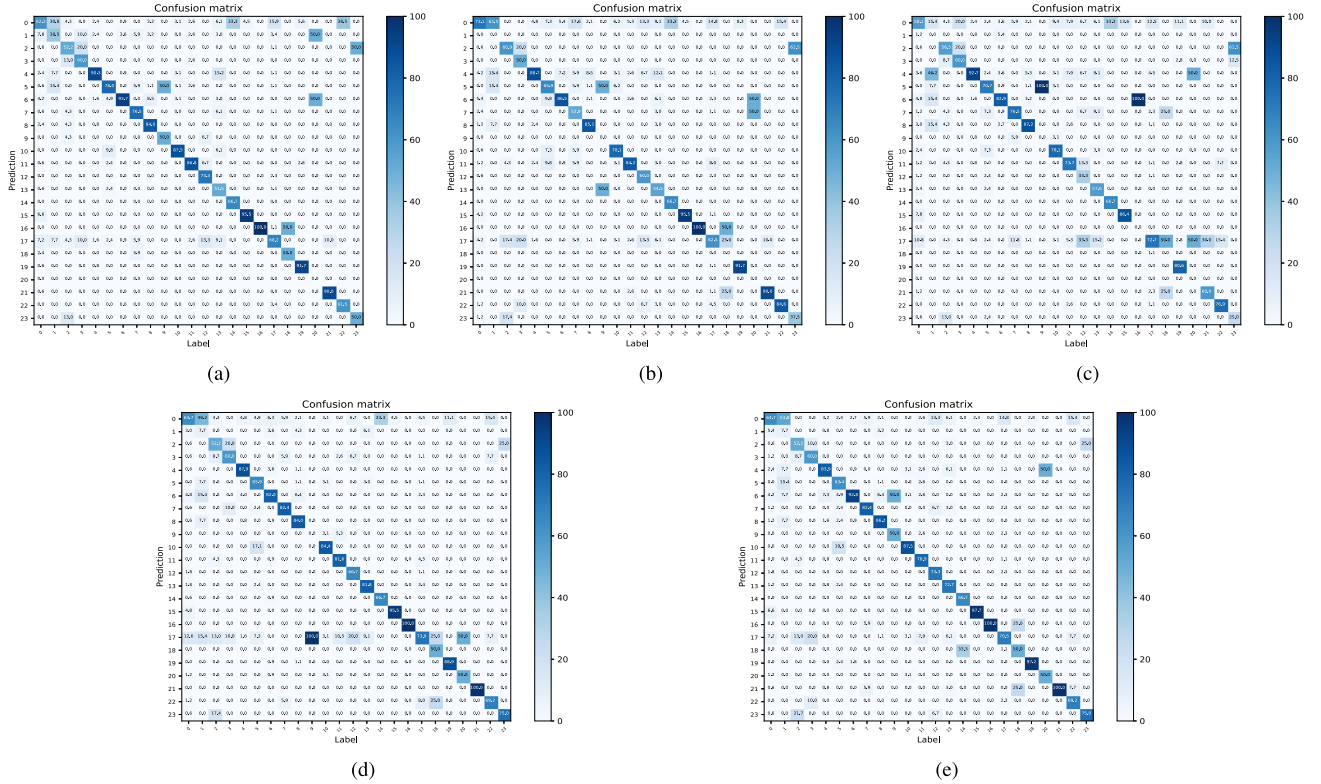


Fig. 6. Confusion matrices of different methods on HRSC2016. The number in the confusion matrix represents the probability that the column's category is predicted as the row's category. (a) LTD. (b) MoEx. (c) PDEN. (d) PMG. (e) Ours.

to take advantage of the M predictions to optimize the student network. MCD takes one step to achieve the optimization orientation, and RCD gradually makes the optimization orientation approach the target optimization orientation. Table V shows the classification results by using MCD and RCD. RCD achieves a better classification performance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
RESULTS OF OUR METHOD (OURS) AND SOTA METHODS ON THE FS23 DATASET. THE OA AND EACH CLASS ACCURACY ARE LISTED

Method	OA	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22
LTD	80.5	87.6	88.2	86.1	77.3	88.1	**77.8**	78.0	94.4	93.5	55.6	89.6	20.0	**93.1**	55.6	**75.0**	85.7	**100.0**	73.9	**75.8**	30.0	61.1	**100.0**	90.9
MoEx	77.2	87.6	64.7	**92.6**	72.7	**94.9**	55.6	76.3	72.2	93.5	66.7	83.3	80.0	75.9	53.3	65.0	85.7	**100.0**	71.0	48.5	10.0	44.4	**100.0**	**100.0**
PDEN	77.7	**93.8**	88.2	84.3	**81.8**	86.4	61.1	**86.4**	83.3	93.5	77.8	85.4	20.0	69.0	37.8	**75.0**	64.3	83.3	73.9	69.7	30.0	44.4	90.0	90.9
PMG	79.0	88.7	88.2	77.8	**81.8**	88.1	55.6	79.7	83.3	**96.8**	72.2	89.6	**100.0**	58.6	48.9	70.0	**100.0**	95.8	75.4	63.6	**50.0**	61.1	**100.0**	90.9
Ours	**83.6**	83.5	**85.3**	84.3	77.3	93.2	**77.8**	**86.4**	**100.0**	**96.8**	**88.9**	**97.9**	60.0	82.8	**60.0**	**75.0**	85.7	**100.0**	**87.0**	72.7	35.0	**66.7**	95.0	**100.0**

TABLE VII
RESULTS OF OUR METHOD (OURS) AND SOTA METHODS ON THE HRSC2016 DATASET. THE OA AND EACH CLASS ACCURACY ARE LISTED

Method	OA	H0	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16	H17	H18	H19	H20	H21	H22	H23
LTD	77.3	62.3	**38.5**	52.2	**60.0**	90.3	**78.0**	**93.7**	76.5	84.0	**50.0**	**87.5**	**86.8**	**73.3**	51.5	**66.7**	95.5	**100.0**	68.2	**50.0**	91.7	0.0	90.0	61.5	50.0
MoEx	75.7	**73.1**	0.0	**60.9**	50.0	88.7	65.9	86.5	52.9	85.1	0.0	78.1	84.2	60.0	54.5	**66.7**	95.5	**100.0**	62.5	0.0	91.7	0.0	90.0	**84.6**	37.5
PDEN	72.7	58.1	0.0	56.5	**60.0**	**92.7**	70.7	82.9	76.5	**87.2**	0.0	78.1	73.7	33.3	57.6	**66.7**	86.4	0.0	72.7	0.0	80.6	0.0	60.0	76.9	25.0
PMG	76.6	64.7	7.7	52.2	**60.0**	87.9	65.9	82.0	**82.4**	84.0	0.0	84.4	81.6	66.7	**81.8**	**66.7**	95.5	**100.0**	**73.9**	**50.0**	88.9	**50.0**	**100.0**	69.2	**75.0**
Ours	**77.5**	64.7	7.7	52.2	**60.0**	83.9	63.4	92.8	**82.4**	86.2	**50.0**	**87.5**	78.9	**73.3**	72.7	**66.7**	**97.7**	**100.0**	70.5	**50.0**	**97.2**	**50.0**	**100.0**	69.2	**75.0**

83.6%, illustrating that optimization orientations gradually approach the target optimization orientation, which can optimize the student network better.

C. Comparison With the SOTA

We compare the proposed DCL model with four SOTA methods, including progressive multigranularity training (PMG) [50], MoEx [11], PDEN [10], and learning-to-diversity (LTD) [51]. We follow the experimental parameter set in the authors' papers to conduct experiments.

1) FS23: Table VI shows the results of our method and the SOTA methods on FS23. Ours achieves an accuracy of 83.6%, the highest accuracy compared with SOTA methods. Specifically, ours improves PMG, MoEx, PDEN, and LTD by 4.6%, 6.4%, 5.9%, and 3.1%, respectively.

2) HRSC2016: We further verify the generalization of our method on the HRSC2016 dataset. The student model is trained using labeled samples in HRSC2016, where the encoder is initialized with the parameters of the one trained on FS23. As shown in Table VII, ours also obtains the highest accuracy of 77.5%.

3) Visualization: Confusion matrices of different methods on FS23 and HRSC2016 are shown in Figs. 5 and 6, respectively. The number in the confusion matrix represents the probability that the column's category is predicted as the row's category. In Fig. 5, our method has a better overall effect than other comparison methods. Note that our method has a high probability of misjudging F19 (Fishing-boat) as F0 (Nonship). The reason may be that F19 has blur details and a similar shape to F0. In Fig. 6, our method has a relatively high predicted value on the diagonal position compared with the other comparison methods. Besides, H9 and H20 are easy to be misclassified due to their too few training samples.

4) Limitation: The proposed model achieves good performance, although it is trained with limited labels. However, its performance degrades when the labeled samples are extremely few. As shown in Table VII, both H18 and H20 obtain 50% accuracy, since their labeled samples are 6 and 3, respectively. Further study may use the sample amplification method to relieve this.

V. CONCLUSION

We attempt to study remote-sensing object recognition with limited labels and present the DCL framework. First, we design the DGM to produce diverse results. DGM is trained with feature diversity loss and cross-entropy loss with labeled samples. Second, we implement the RCDM to distill the diverse knowledge from the DGM to the student network. Especially, this process uses unlabeled samples. Extensive experiments are built on two widely used datasets of FS23 and HRSC2016 that verify the superior performance of our method.

REFERENCES

[1] W. Zhao et al., "Feature balance for fine-grained object classification in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

[2] W. Zhao, J. Liu, Y. Liu, F. Zhao, Y. He, and H. Lu, "Teaching teachers first and then student: Hierarchical distillation to improve long-tailed object recognition in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[3] X. Hu, P. Zhang, and Q. Zhang, "A novel framework of CNN integrated with AdaBoost for remote sensing scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 2643–2646.

[4] Y. Wan, Y. Zhong, A. Ma, J. Wang, and R. Feng, "RSSM-Net: Remote sensing image scene classification based on multi-objective neural architecture search," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 1369–1372.

[5] X. Zhang, X. Tan, G. Chen, K. Zhu, P. Liao, and T. Wang, "Object-based classification framework of remote sensing images with graph convolutional networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[6] J. Wang, Y. Zhong, Z. Zheng, A. Ma, and L. Zhang, "RSNet: The search for remote sensing deep neural networks in recognition tasks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2520–2534, Mar. 2020.

[7] X. Zhang, Y. Lv, L. Yao, W. Xiong, and C. Fu, "A new benchmark and an attribute-guided multilevel feature representation network for fine-grained ship classification in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1271–1285, 2020.

[8] C. M. Ward, J. Harguess, and C. Hilton, "Ship classification from overhead imagery using synthetic data and domain adaptation," in *Proc. OCEANS*, Oct. 2018, pp. 1–5.

[9] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods*, vol. 2, 2017, pp. 324–331.

[10] L. Li et al., "Progressive domain expansion network for single domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 224–233.

[11] B. Li, F. Wu, S.-N. Lim, S. Belongie, and K. Q. Weinberger, "On feature normalization and data augmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12383–12392.

[12] Z. Hu, Z. Yang, X. Hu, and R. Nevatia, "SimPLE: Similar pseudo label exploitation for semi-supervised classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15099–15108.

[13] J.-C. Su, Z. Cheng, and S. Maji, "A realistic evaluation of semi-supervised learning for fine-grained classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 12966–12975.

[14] W. Lin, Z. Gao, and B. Li, "Shoestring: Graph-based semi-supervised classification with severely limited labeled data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4174–4182.

[15] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," 2020, *arXiv:2001.07685*.

[16] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 2613–2622.

[17] X. Lai et al., "Semi-supervised semantic segmentation with directional context-aware consistency," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 1205–1214.

[18] X. Huo et al., "ATSO: Asynchronous teacher-student optimization for semi-supervised image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 1235–1244.

[19] J. Lee, E. Kim, and S. Yoon, "Anti-adversarially manipulated attributions for weakly and semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 4071–4080.

[20] L. Chen, T. Yang, X. Zhang, W. Zhang, and J. Sun, "Points as queries: Weakly semi-supervised object detection by points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8823–8832.

[21] Z. Wang, Y. Li, Y. Guo, L. Fang, and S. Wang, "Data-uncertainty guided multi-phase learning for semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4568–4577.

[22] Y. Tang, W. Chen, Y. Luo, and Y. Zhang, "Humble teachers teach better students for semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3132–3141.

[23] H. Wang, Y. Cong, O. Litany, Y. Gao, and L. J. Guibas, "3DIoUMatch: Leveraging IoU prediction for semi-supervised 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 14615–14624.

[24] J. Jeong, V. Verma, M. Hyun, J. Kannala, and N. Kwak, "Interpolation-based semi-supervised learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 11602–11611.

[25] Q. Zhou, C. Yu, Z. Wang, Q. Qian, and H. Li, "Instant-teaching: An end-to-end semi-supervised object detection framework," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4081–4090.

[26] R. Valle, J. M. Buenaposada, A. Valdes, and L. Baumela, "A deeply-initialized coarse-to-fine ensemble of regression trees for face alignment," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 585–601.

[27] W. Kim, B. Goyal, K. Chawla, J. Lee, and K. Kwon, "Attention-based ensemble for deep metric learning," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 736–751.

[28] N. Dvornik, C. Schmid, and J. Mairal, "Diversity with cooperation: Ensemble methods for few-shot classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 3723–3731.

[29] N. Aziere and S. Todorovic, "Ensemble deep manifold similarity learning using hard proxies," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7299–7307.

[30] R. Mackowiak, L. Ardizzone, U. Kothe, and C. Rother, "Generative classifiers as a basis for trustworthy image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2971–2981.

[31] R. Ji et al., "Attention convolutional binary neural tree for fine-grained visual categorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10468–10477.

[32] P. Wang, K. Han, X.-S. Wei, L. Zhang, and L. Wang, "Contrastive learning based hybrid networks for long-tailed image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 943–952.

[33] J. Ma et al., "Remote sensing scene classification based on global and local consistent network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 537–540.

[34] C. Zhang, B. Xiong, and G. Kuang, "Ship detection and recognition in optical remote sensing images based on scale enhancement rotating cascade R-CNN networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3545–3548.

[35] J. Chen and Y. Qian, "Hierarchical multi-label ship recognition in remote sensing images using label relation graphs," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 4968–4971.

[36] Q. Yang, X. Wei, B. Wang, X.-S. Hua, and L. Zhang, "Interactive self-training with mean teachers for semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5941–5950.

[37] D.-H. Lee et al., "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *Proc. Workshop Challenges Represent. Learn. (ICML)*, vol. 3, no. 2, 2013, p. 896.

[38] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12674–12684.

[39] S. Laine and T. Aila, "Temporal ensembling for semi-supervised learning," 2016, *arXiv:1610.02242*.

[40] D. Berthelot et al., "ReMixMatch: Semi-supervised learning with distribution alignment and augmentation anchoring," 2020, *arXiv:1911.09785*.

[41] T. Miyato, S.-I. Maeda, M. Koyama, and S. Ishii, "Virtual adversarial training: A regularization method for supervised and semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1979–1993, Aug. 2018.

[42] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," 2017, *arXiv:1703.01780*.

[43] W. Zhao, X. Hou, Y. He, and H. Lu, "Defocus blur detection via boosting diversity of deep ensemble networks," *IEEE Trans. Image Process.*, vol. 30, pp. 5426–5438, 2021.

[44] K. Meshgi, S. Oba, and S. Ishii, "Efficient diverse ensemble for discriminative co-tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4814–4823.

[45] A. Vyas, N. Jammalamadaka, X. Zhu, D. Das, B. Kaul, and T. L. Willke, "Out-of-distribution detection using an ensemble of self supervised leave-out classifiers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 550–564.

[46] W. Zhao, B. Zheng, Q. Lin, and H. Lu, "Enhancing diversity of defocus blur detectors via cross-ensemble network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8905–8913.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.

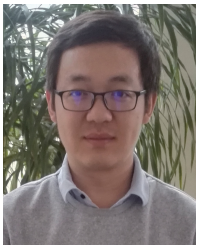
[47] Z. Shi et al., "Crowd counting with deep negative correlation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5382–5390.

[48] L. Zhang et al., "Nonlinear regression via deep negative correlation learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 3, pp. 982–998, Mar. 2019.

[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

[50] R. Du et al., "Fine-grained visual classification via progressive multi-granularity training of jigsaw patches," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 153–168.

[51] Z. Wang, Y. Luo, R. Qiu, Z. Huang, and M. Baktashmotlagh, "Learning to diversify for single domain generalization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 834–843.



Wenda Zhao (Member, IEEE) received the B.S. degree from Jilin University, Changchun, China, in 2011, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the School of Information and Communication Engineering, Dalian University of Technology (DUT), Dalian, China. His research interests include deep learning and computer vision tasks.



Tingting Tong received the B.S. degree from the North China University of Water Resources and Electric Power, Zhengzhou, China, in 2019, and the M.S. degree from the Dalian University of Technology, Dalian, China, in 2022.

Her research interests include object classification.



Haipeng Wang received the Ph.D. degree from Naval Aviation University, Yantai, China, in 2012.

He is currently a Professor with Naval Aviation University. His research interests include the general area of intelligent perception and fusion, and big data technology and application.



Fan Zhao received the B.S. degree from the Changchun University of Science and Technology, Changchun, China, in 2011, and the M.S. and Ph.D. degrees from the Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, in 2014 and 2017, respectively.

She is currently a Faculty Member with the School of Physics and Electronic Technology, Liaoning Normal University, Dalian, China. Her research interests include low-level and high-level vision tasks.



You He received the Ph.D. degree from Tsinghua University, Beijing, China, in 1997.

He was selected as the Chinese Academy of Engineering Academician in 2013. He is currently a Professor with the Department of Electronic Engineering, Tsinghua University; the Director of the Institute of Information Fusion, Naval Aviation University, Yantai, China, where he is the Director of the Key Laboratory of Information Perception and Fusion Technology. His main research interests include information fusion, computer vision, and big data technology.

Dr. He is an IET Fellow. He received four second prizes of the National Science and Technology Progress and seven first prizes of the Military Science and Technology Progress.



Huchuan Lu (Senior Member, IEEE) received the M.S. degree in signal and information processing and the Ph.D. degree in system engineering from the Dalian University of Technology (DUT), Dalian, China, in 1998 and 2008, respectively.

He has been a Faculty Member since 1998 and a Professor since 2012 with the School of Information and Communication Engineering, DUT. His research interests include computer vision and pattern recognition. In recent years, he focuses on visual tracking, detection, and segmentation.

Dr. Lu currently serves as an Associate Editor of the IEEE TRANSACTIONS ON CYBERNETICS and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:21:50 UTC from IEEE Xplore. Restrictions apply.